

JEE MAIN REVISION MASTER: HYDROGEN OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 86 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Hydrogen — Core Principles & Governing Laws	47 PYQs	Core Principle
Very High	Hydrogen — Applied Cases, Reactions & Traps	45 PYQs	Applied Topics
Very High	Hydrogen — Standard Calculations & Problem Archetypes	22 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hydrogen — Core Principles & Governing Laws	$2\text{Mn}^{+7}\text{O}_4^{2-} + 3\text{H}_2\text{O}_2 \rightarrow \text{NaH} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2, E_{\text{H}}^{\text{D}} - 7.5$
Hydrogen — Applied Cases, Reactions & Traps	$2\text{Mn}^{+7}\text{O}_4^{2-} + 3\text{H}_2\text{O}_2 \rightarrow \text{Ni}^{1270\text{K}}, 2\text{MnO}_4^{2-} + 6\text{H}^{+} + 5\text{H}_2\text{O}_2 \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 5\text{O}_2$
Hydrogen — Standard Calculations & Problem Archetypes	$\text{O}_2, \text{CaCl}_2, \text{CaSO}_4, \text{MgCl}_2, \text{H}_2\text{O}_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrogen — Core Principles & Governing Laws (Very High Yield — 47 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Hydrogen.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Hydrogen.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Hydrogen PYQs.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Isotopes of hydrogen hav..." [Ans: B]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: Statement-I : Methane and steam passed over a heated Ni catalyst produces hydrogen gas. Statement-II : Sod..." [Ans: C]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Loss of electron from hy..." [Ans: N/A]

3.2 Hydrogen — Applied Cases, Reactions & Traps (Very High Yield — 45 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Hydrogen.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Isotopes of hydrogen hav..." [Ans: B]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements: Statement-I : Methane and steam passed over a heated Ni catalyst produces hydrogen gas. Statement-II : Sod..." [Ans: C]
- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : H_2O_2 is used in the synthesis of Cephalosporin Statement II : H_2O_2 is used fo..." [Ans: N/A]

3.3 Hydrogen — Standard Calculations & Problem Archetypes (Very High Yield — 22 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Hydrogen.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : H_2O_2 is used in the synthesis of Cephalosporin Statement II : H_2O_2 is used fo..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : Physical properties o..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "The starting material for convenient preparation of deuterated hydrogen peroxide (D_2O_2) in ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 13th April Evening Shift (Q#1)

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Isotopes of hydrogen have almost same chemical properties, but difference in their rates of reaction. Reason R : Isotopes of hydrogen have different enthalpy of bond dissociation. In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) is not correct but R is correct
- (B) Both A and R are correct but R is NOT the correct explanation of A [**Correct**]
- (C) is correct but R is not correct
- (D) Both A and R are correct and R is the correct explanation of A

Correct Answer: B

Solution Analysis: Assertion A : Isotopes of hydrogen (protium, deuterium, and tritium) have almost the same chemical properties because they have the same number of protons and electrons, which dictate chemical behavior. However, they differ in their rates of reaction due to differences in their atomic masses, which affect reaction kinetics. Reason R : The isotopes of hydrogen have different enthalpy of bond dissociation due to their differences in atomic mass. The stronger bond in deuterium compared to protium, for example, leads to higher bond dissociation enthalpy. This, in turn, influences the rate of react

Benchmark Problem 2: JEE Main 2023 (Online) 31st January Evening Shift (Q#1)

Given below are two statements : Statement I : H_2O_2 is used in the synthesis of Cephalosporin Statement II : H_2O_2 is used for the restoration of aerobic conditions to sewage wastes. In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Statement I is incorrect but Statement II is correct
- (B) Statement I is correct but Statement II is incorrect
- (C) Both Statement I and Statement II are incorrect
- (D) Both Statement I and Statement II are correct

Correct Answer: None

Solution Analysis: H_2O_2 is used in the synthesis of hydroquinone, tartaric acid and certain food products and pharmaceuticals (cephalosporin). Nowadays it is also used in environmental (green) chemistry for example in pollution control treatment of domestic and industrial effluents, oxidation of cyanides restoration of aerobic condition to sewage waste. Hence both statements are correct.

Benchmark Problem 3: JEE Main 2023 (Online) 13th April Morning Shift (Q#2)

Given below are two statements : Statement I : Permutit process is more efficient compared to the synthetic resin method for the softening of water. Statement II : Synthetic resin method results in the formation of soluble sodium salts. In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Statement I is correct but Statement II is incorrect
- (B) Both the Statements I and II are correct
- (C) Both the Statements I and II are incorrect
- (D) Statement I is incorrect but Statement II is correct

Correct Answer: None

Solution Analysis: I. Synthetic resin is more efficient as cations as well as anions responsible for removal of permanent hardness. II. In synthetic resin method, cations and anions are removed in the form of ppt. Hence, both statements are incorrect.